t3-w04 — condition C3

Three Dimensional Shapes — Three dimensional figures: properties of cubes, cuboids, pyramids and cones

> C2 plus a validation and repair pass over the draft.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	**present**

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

entry_behaviour

• Students have been taught the differences between cube and cuboid.

instructional_materials

• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.

Objective 1 — Identify the properties of cube and cuboids

objective_text

Identify the properties of cube and cuboids

current_step

1

instructional_events

• present_content

assessment_points

• How many vertices, has a cuboid?
• How many edges has a cube?
• Draw a cube and cuboid then label.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Three Dimensional shapes\n\nThese are shapes that have length, breadth and
height. They are called solid shapes.\n\nThey may be either regular or irregular shapes. Regular
shapes are cube of sugar, can drinks,\nmatch box, etc. Irregular shapes are stone, trees etc.
\n\n#### Properties of cube\n\nA cube drawn as a see-through solid: two equal squares, the nearer
one offset down and to the\nright of the farther one, with the four pairs of corresponding
corners joined.\n\n[drawing attached to this section here, unlabelled]\n\n1. It is a solid
state\n2. It has 6 square faces\n3. It has 12 edges\n4. It has 8 vertices.\n\n**Hint.** An edge
is a line where two faces meet. It may be straight or curved. A face or\nsurface may be flat or
curved. A vertex is a point corner where edges meet.\n\n#### Properties of a Cuboid\n\nA cuboid
drawn the same way, wide and shallow, with three leader lines to the right naming the\ntop face,
the nearer top-right corner and the nearer right-hand vertical edge.\n\n[drawing attached to this
section here, unlabelled, marked face, vertex, edge]\n\n1. It has 6 faces\n2. It has 12 edges\n3.
It has 8 vertices."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

`current_step`

2

`instructional_events`

• present_content

`assessment_points`

• Draw an hexagonal base pyramid and list the properties.
• How many vertices has a tetrahedron.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Pyramids\n\nA pyramid is named based on its base shape.\n\nA triangular base
pyramid. The base triangle is drawn in perspective, its third vertex raised\ninto the middle of
the figure, and the apex above is joined to all three base vertices.\n\n[drawing attached to this
section here, unlabelled, marked Triangular base Pyramid (Tetrahedron)]\n\nA square base pyramid.
The base is drawn as an upright rectangle and the apex above is joined\nto all four of its
corners, so two edges run outside the rectangle and two cross it.\n\n[drawing attached to this
section here, unlabelled, marked Square base Pyramid]\n\nA pentagonal based pyramid. The base is
drawn as a pentagon standing on one side with its\nfifth vertex uppermost, and the apex above is
joined to that vertex, to the two vertices\neither side of it, and to the lower-right vertex —
four edges from the apex, the last of them\ncrossing the face of the solid.\n\n[drawing attached
to this section here, unlabelled, marked Pentagonal based Pyramid]\n\nA rectangular based
pyramid, drawn like the square one but on a base wider than it is tall.\n\n[drawing attached to
this section here, unlabelled, marked Rectangular based Pyramid]\n\n##### Properties\n\n| Shape |
Faces | edges | Vertices |\n|---|---|---|---|\n| Triangular based Pyramid | 4 | 6 | 4 |\n| Square
based Pyramid | 5 | 8 | 5 |\n| Pentagonal base Pyramid | 6 | 10 | 6 |\n| Rectangular based
Pyramid | 5 | 8 | 5 |\n\n#### Cone\n\nTwo drawings side by side. On the left an upright cone: an
elliptical base with the two slant\nsides running up from the ends of the ellipse to the apex. On
the right a cone lying on its\nside, apex at the left, its two slant sides reaching a narrow
upright ellipse whose top and\nbottom then continue as two parallel lines to a second narrow
ellipse further right.\n\n[drawing attached to this section here, unlabelled]\n\n1. It has two
faces, i.e. one curved face and one flat face.\n2. It has one edge.\n3. It has one vertex"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Properties of Cubes and Cuboids

EXPLANATION

A **cube** is a solid shape with three equal dimensions. It has:
- 6 square faces (each face is the same size)
- 12 edges (the line where two faces meet)
- 8 vertices (the corner points where edges meet)

A **cuboid** is similar to a cube but its length, width and height can be different. It still has the same number of parts as a cube:
- 6 faces (rectangles, not necessarily squares)
- 12 edges
- 8 vertices

Remember: an **edge** is a line where two faces join, and a **vertex** is a corner where edges meet.

WORKED_EXAMPLE

Worked example:

Count the edges of a cube.

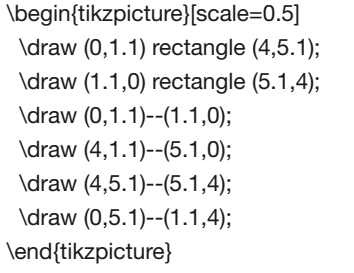
1. Look at one face of the cube – it has 4 edges.
2. The cube has 6 faces, but each edge belongs to two faces, so counting each face's edges would double-count.
3. Use the formula for a rectangular solid: $\text{edges} = 12$.

Thus a cube has **12 edges**.

Now count the vertices of a cuboid.

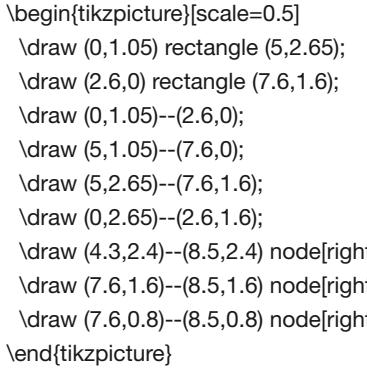
1. Each corner of the top face is a vertex, giving 4 vertices.
2. The bottom face also has 4 vertices.
3. No vertex is shared between top and bottom, so total vertices = 4 + 4 = **8 vertices**.

VISUAL



```
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.1) rectangle (4,5.1);
  \draw (1.1,0) rectangle (5.1,4);
  \draw (0,1.1)--(1.1,0);
  \draw (4,1.1)--(5.1,0);
  \draw (4,5.1)--(5.1,4);
  \draw (0,5.1)--(1.1,4);
\end{tikzpicture}
```

VISUAL



```
\begin{tikzpicture}[scale=0.5]
  \draw (0,1.05) rectangle (5,2.65);
  \draw (2.6,0) rectangle (7.6,1.6);
  \draw (0,1.05)--(2.6,0);
  \draw (5,1.05)--(7.6,0);
  \draw (5,2.65)--(7.6,1.6);
  \draw (0,2.65)--(2.6,1.6);
  \draw (4.3,2.4)--(8.5,2.4) node[right] {face};
  \draw (7.6,1.6)--(8.5,1.6) node[right] {vertex};
  \draw (7.6,0.8)--(8.5,0.8) node[right] {edge};
\end{tikzpicture}
```

1. How many vertices does a cuboid have?

answer: 8

2. How many edges does a cube have?

answer: 12

3. How many faces does a cube have, and what shape are they?

answer: 6 square faces

Properties of Pyramids and Cones

EXPLANATION

A **pyramid** is a solid that has a polygonal base and triangular faces that meet at a single point called the apex. The name of a pyramid tells you the shape of its base:

- **Triangular (tetrahedron)** – base is a triangle.
- **Square** – base is a square.
- **Rectangular** – base is a rectangle (different length and width).
- **Pentagonal** – base is a pentagon.

For each type, count the faces, edges and vertices:

Pyramid type	Faces (including the base)	Edges	Vertices
Triangular (tetrahedron)	4	6	4
Square	5	8	5
Rectangular	5	8	5
Pentagonal	6	10	6

A **cone** has a circular (or elliptical) base and a single curved surface that meets at the apex. It therefore has:

- 2 faces – one curved face and one flat circular (or elliptical) face.
- 1 edge – the circular edge where the base meets the curved surface.
- 1 vertex – the apex.

These properties let you identify any pyramid or cone you see.

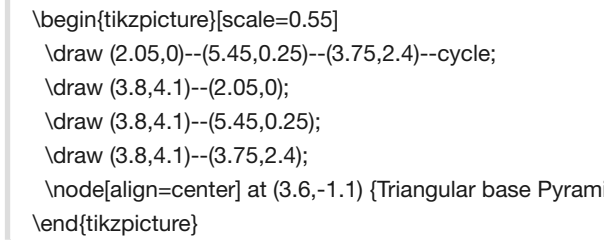
WORKED_EXAMPLE

Worked Example – Counting the parts of a square-based pyramid

1. Look at the solid: it has a square base and four triangular faces that meet at the apex.
2. **Faces:** The base counts as one face, and each triangular side is a face. So 1 (base) + 4 (triangles) = 5 faces.
3. **Edges:** The base has 4 edges. Each triangle adds one edge from a base corner up to the apex, giving another 4 edges. Total edges = 4 + 4 = 8.
4. **Vertices:** The base provides 4 corner vertices, and the apex adds one more. Total vertices = 4 + 1 = 5.

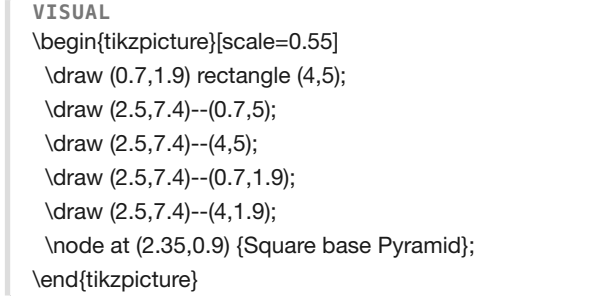
Thus a square-based pyramid has 5 faces, 8 edges, and 5 vertices, matching the table above.

VISUAL




```
\begin{tikzpicture}[scale=0.55]
  \draw (2.05,0)--(5.45,0.25)--(3.75,2.4)--cycle;
  \draw (3.8,4.1)--(2.05,0);
  \draw (3.8,4.1)--(5.45,0.25);
  \draw (3.8,4.1)--(3.75,2.4);
  \node[align=center] at (3.6,-1.1) {Triangular base Pyramid\\(Tetrahedron)};
\end{tikzpicture}
```

VISUAL




```
\begin{tikzpicture}[scale=0.55]
  \draw (0.7,1.9) rectangle (4,5);
  \draw (2.5,7.4)--(0.7,5);
  \draw (2.5,7.4)--(4,5);
  \draw (2.5,7.4)--(0.7,1.9);
  \draw (2.5,7.4)--(4,1.9);
  \node at (2.35,0.9) {Square base Pyramid};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.55]
  \draw (3.1,6.75)--(5.3,4.5)--(4.15,1.2)--(1.65,1.25)--(1.2,4.4)--cycle;
```

```
  \draw (3.3,9.65)--(1.2,4.4);
  \draw (3.3,9.65)--(3.1,6.75);
  \draw (3.3,9.65)--(5.3,4.5);
  \draw (3.3,9.65)--(4.15,1.2);
  \node[align=center] at (3,0.1) {Pentagonal based\\Pyramid};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.55]
  \draw (1.15,2.3) rectangle (5,4.45);
  \draw (2.9,8)--(1.15,4.45);
  \draw (2.9,8)--(5,4.45);
  \draw (2.9,8)--(1.15,2.3);
  \draw (2.9,8)--(5,2.3);
  \node[align=center] at (3,1.2) {Rectangular based\\Pyramid};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.5]
  \draw (0,0) ellipse (1.6 and 0.83);
  \draw (-1.6,0)--(-0.15,4.1)--(1.6,0);
  \begin{scope}[xshift=7cm]
    \draw (2.6,1)--(0,0)--(2.6,-1);
    \draw (2.6,0) ellipse (0.45 and 1);
    \draw (9.6,0) ellipse (0.5 and 1.05);
    \draw (2.6,1)--(9.6,1.05);
    \draw (2.6,-1)--(9.6,-1.05);
  \end{scope}
\end{tikzpicture}
```

1. 1. List the number of faces, edges and vertices of a triangular (tetrahedron) pyramid.
answer: Faces: 4, Edges: 6, Vertices: 4

2. 2. How many edges does a cone have?
answer: 1 edge

3. 3. A pyramid has a pentagonal base. How many vertices does it have?
answer: 6 vertices

4. 4. Draw a rectangular based pyramid and write down its number of faces, edges and vertices.
answer: Faces: 5, Edges: 8, Vertices: 5

5. 5. How many faces does a cone have?
answer: 2 faces

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Enacted content	**grounded**
Worked example a problem with its solution worked through	2	Enacted content	**grounded**
Figure a drawing the learner reads from	7	Figures	**grounded**
	8	Assessment	**substituted**

Practice what the learner does		written from the teacher's evaluation questions instead of her set work	
Answer what the learner checks against	8	Enacted content	**grounded**